GENOMICS AND BIOTECHNOLOGY

Module map

- Module 11: Genomics and Biotechnology
- Connected lab: lab-11_genomics-biotechnology.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 11

Genomics and Biotechnology

1 Genomes as data

2 PCR and amplification

3 Gel electrophoresis and comparison

4 Genetic engineering tools

5 Biotechnology decisions and ethics

Lab evidence

lab-11_genomics-biotechnology.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

GENOMICS AND BIOTECHNOLOGY

Learning objectives

- Define genome and genomics.
- Explain why PCR amplification is useful.
- Interpret basic gel electrophoresis fragment patterns.
- Describe what genetic engineering tools can do.
- Evaluate biotechnology claims using evidence and ethical considerations.

OBJECTIVE 1
Define genome and genomics.

OBJECTIVE 2
Explain why PCR amplification is useful.

OBJECTIVE 3
Interpret basic gel electrophoresis fragment patterns.

OBJECTIVE 4
Describe what genetic engineering tools can do.

OBJECTIVE 5
Evaluate biotechnology claims using evidence and ethical considerations.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

GENOMICS AND BIOTECHNOLOGY

Topic sequence

- Genomes as data: A genome is the full DNA information set of an organism or sample.
- PCR and amplification: PCR amplifies chosen DNA regions so they can be analyzed.
- Gel electrophoresis and comparison: Gel electrophoresis separates DNA fragments by size.
- Genetic engineering tools: Biotechnology tools can copy, edit, compare, or move genetic information.
- Biotechnology decisions and ethics: Applications require evidence, risk reasoning, and ethical judgment.

01

Genomes as data

A genome is the full DNA information set of an organism or sample.

02

PCR and amplification

PCR amplifies chosen DNA regions so they can be analyzed.

03

Gel electrophoresis and comparison

Gel electrophoresis separates DNA fragments by size.

04

Genetic engineering tools

Biotechnology tools can copy, edit, compare, or move genetic information.

05

Biotechnology decisions and ethics

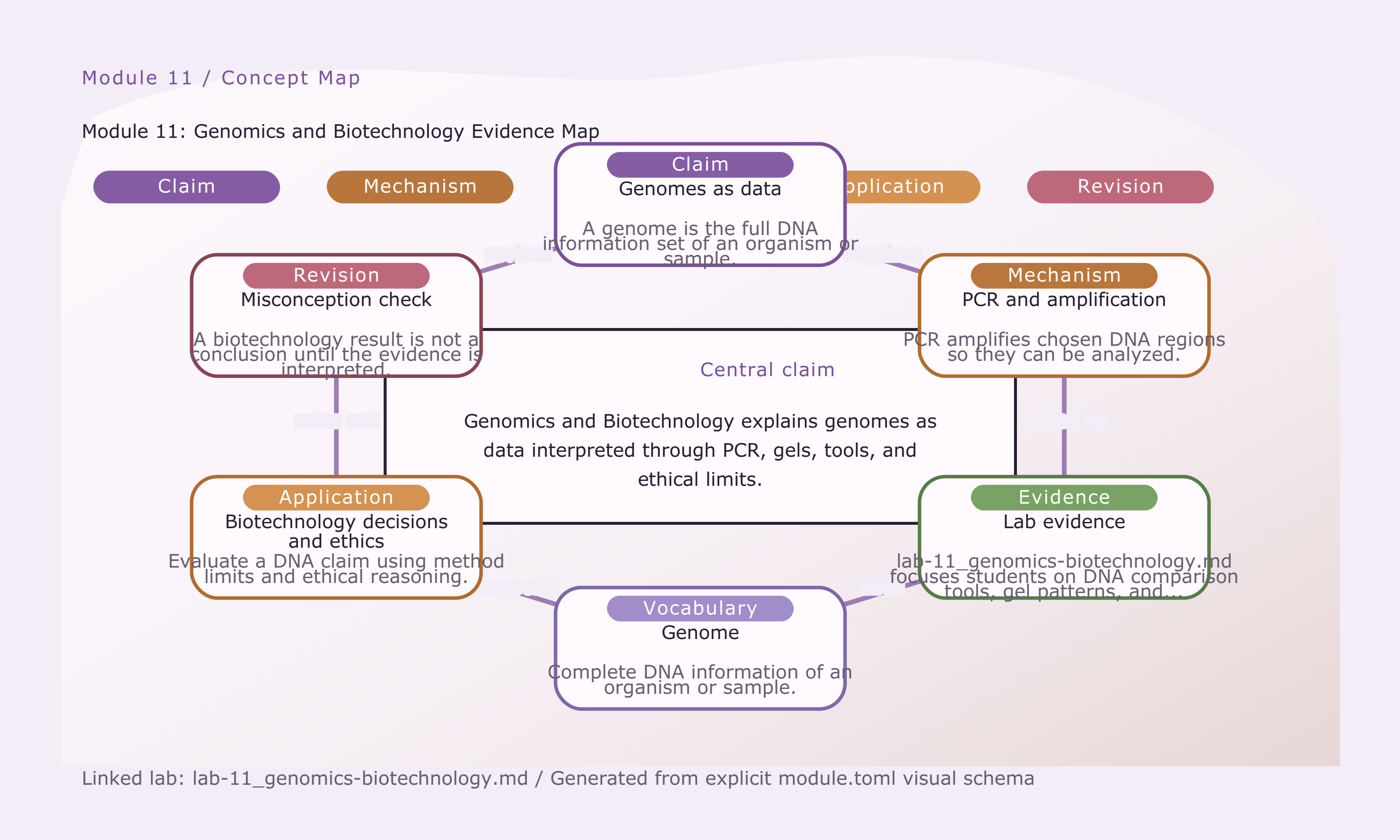
Applications require evidence, risk reasoning, and ethical judgment.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

GENOMICS AND BIOTECHNOLOGY

Module 11: Genomics and Biotechnology Evidence Map

- Module focus: Genomics and Biotechnology
- Central claim: Genomics and Biotechnology explains genomes as data interpreted through PCR, gels, tools, and ethical limits.
- Key nodes to connect: Genomes as data, PCR and amplification, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-11_genomics-biotechnology.md supplies an evidence surface for the map.

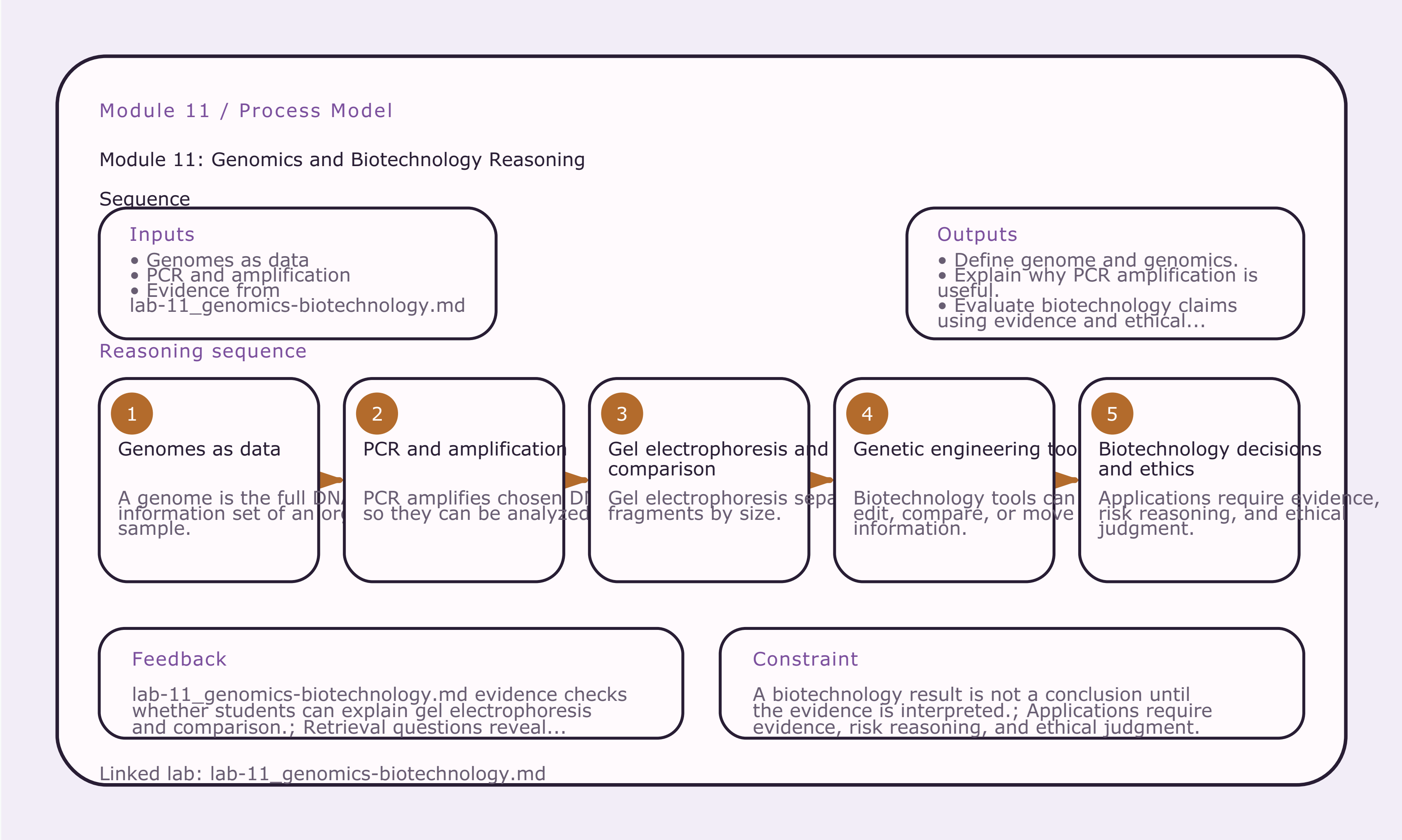


Teaching note: Ask students to explain one edge in their own words before moving on.

GENOMICS AND BIOTECHNOLOGY

Module 11: Genomics and Biotechnology Reasoning Sequence

- Module focus: Genomics and Biotechnology
- Trace the model: Genomes as data -> PCR and amplification -> Gel electrophoresis and comparison -> Genetic engineering tools
- Inputs become outputs: Genomes as data, PCR and amplification -> Define genome and genomics., Explain why PCR amplification is useful.
- Feedback check: lab-11_genomics-biotechnology.md evidence checks whether students can explain gel electrophoresis and comparison.
- Lab connection: lab-11_genomics-biotechnology.md gives students a process to observe or test.



Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

GENOMICS AND BIOTECHNOLOGY

Terms as evidence handles

- Genome: Complete DNA information of an organism or sample.
- Genomics: Study of whole genomes and their patterns.
- PCR: Technique for amplifying selected DNA sequences.
- Primer: Short DNA sequence that starts PCR copying.
- Gel electrophoresis: Method that separates DNA fragments by size.
- CRISPR: Gene-editing system adapted from bacterial defense.

GENOME
Complete DNA information of an organism or sample.

GENOMICS
Study of whole genomes and their patterns.

PCR
Technique for amplifying selected DNA sequences.

PRIMER
Short DNA sequence that starts PCR copying.

GEL ELECTROPHORESIS
Method that separates DNA fragments by size.

CRISPR
Gene-editing system adapted from bacterial defense.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

GENOMICS AND BIOTECHNOLOGY

Lab connection

- Lab file: lab-11_genomics-biotechnology.md
- Lab evidence should test or illustrate: Gel electrophoresis and comparison
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Genomes as data

EVIDENCE
lab-11_genomics-biotechnology.md

CLAIM
Define genome and genomics.

Teaching note: Make the lab connection explicit before students encounter the activity.

GENOMICS AND BIOTECHNOLOGY

Contrast check

- Do not stop at: A genome is the full DNA information set of an organism or sample.
- Stronger explanation: Applications require evidence, risk reasoning, and ethical judgment.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
A genome is the full DNA information set of an organism or sample.

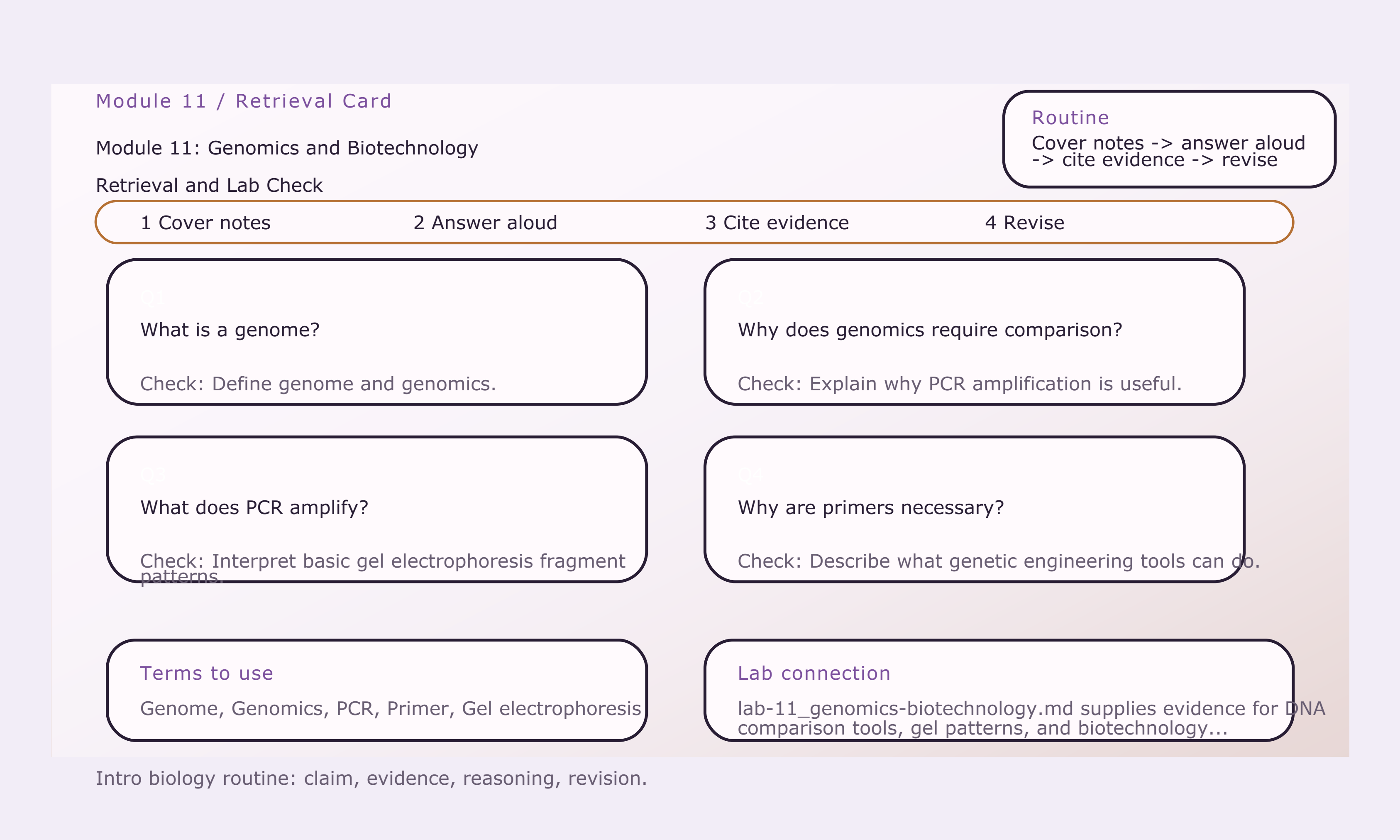
DEEPER
Applications require evidence, risk reasoning, and ethical judgment.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

GENOMICS AND BIOTECHNOLOGY

Module 11: Genomics and Biotechnology Retrieval and Lab Check

- Module focus: Genomics and Biotechnology
- Retrieval prompt: What is a genome?
- Answer check: Define genome and genomics.
- Required terms: Genome, Genomics, PCR
- Lab connection: lab-11_genomics-biotechnology.md; lab-11_genomics-biotechnology.md supplies evidence for DNA comparison tools, gel patterns, and biotechnology decisions.



Teaching note: Pause here. Students answer first without notes, then compare to the checks.

GENOMICS AND BIOTECHNOLOGY

Practice quiz bridge

- 1. What does PCR do?
- 2. Gel electrophoresis separates DNA fragments by:
- 3. CRISPR is best described as a tool that can:
- 4. Why does biotechnology require bioethics?

QUESTION 1

Key: C

What does PCR do?

QUESTION 2

Key: A

Gel electrophoresis separates DNA fragments by:

QUESTION 3

Key: D

CRISPR is best described as a tool that can:

QUESTION 4

Key: B

Why does biotechnology require bioethics?

Teaching note: Use the quiz as formative feedback. Answer key: 1=C, 2=A, 3=D, 4=B

GENOMICS AND BIOTECHNOLOGY

Synthesis and exit ticket

- Exit claim: explain genomes as data using evidence from lab-11_genomics-biotechnology.md.
- Must include: Genome, Genomics, PCR.
- Revision prompt: what evidence would change your explanation?

CLAIM
Genomes as data

EVIDENCE
lab-11_genomics-biotechnology.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.